45 Posttransplant Lymphoproliferative Syndromes

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45.1 Definitions

Posttransplant lymphoproliferative disorders (PTLD) constitute a heterogeneous group of lymphoproliferative diseases that occur in the setting of transplantation and result from the uncontrolled neoplastic proliferation of lymphoid or plasmacytic cells in the context of extrinsic immunosuppression after transplantation.

PTLD in the HCT setting are largely caused by latent Epstein-Barr virus (EBV, HHV-4), belonging to the herpesviruses family. It is one of the most common viruses in humans, with prevalence of 82–84% in the overall population (Styczynski et al. 2016a). EBV is associated with the development of various diseases, which can be categorized as primary syndromes, EBV-associated tumors, and EBV-associated posttransplant diseases: PTLD and other end-organ diseases (encephalitis/myelitis, pneumonitis, hepatitis, or hemophagocytic lymphohistiocytosis).

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Comparably to other herpersviruses, there are two types of EBV infection: primary and recurrent. Primary EBV infection is diagnosed when EBV is detected (nucleic acid or serologically) in an EBV-naïve individual. Recurrent EBV-DNA-emia (previously: latent infection) is diagnosed by detection of EBV-DNA in the blood, in a previously infected individual (Dharnidharka et al. 2016).

45.2 Types of PTLD

PTLD or end-organ EBV-associated posttransplant disease can be diagnosed at the probable or proven level. Probable EBV disease is diagnosed in case of significant lymphadenopathy, hepatosplenomegaly, or other end-organ manifestations (without tissue biopsy, but in the absence of other documented cause) together with high EBV-DNA-emia. Proven EBV disease (PTLD or other end-organ disease) is diagnosed in case of symptoms and/or signs from the affected organ together with the detection of EBV-encoded RNA by in situ hybridization (EBER-ISH) in a tissue specimen (immunohistochemistry for EBV proteins have good specificity but lower sensitivity; these proteins are variably expressed in PTLD biopsies). Histological WHO 2016 classification included six types of morphological PTLD: plasmacytic hyperplasia, infectious mononucleosis-like, florid follicular hyperplasia, polymorphic, monomorphic (B-cell or T-/NK-cell types), and

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A. Sureda et al. (eds.), *The EBMT Handbook*, https://doi.org/10.1007/978-3-031-44080-9_45

classical Hodgkin lymphoma PTLD. In the 2022 WHO classification of lymphoid malignancies, PTLD are not any longer listed and are defined as immunodeficiency-associated lymphoproliferative disorders (Marjanska et al. 2023).

Atypical PTLD

- EBV-negative PTLD: A growing number of cases of EBV-negative PTLD have been reported, mainly in SOT recipients. These cases tend to present later (>5 years) after transplant, and an increased risk is observed as long as 10 years after transplantation. These cases should be regarded as malignant lymphoma rather than as PTLD.
- T-lineage PTLD: T-PTLD is usually EBV-negative and the relatively long latency between transplantation and T-PTLD onset may be explained by molecular events. The frequency of T-PTLDs ranges 4–15% of all PTLD cases. EBV is present in approximately one-third of T-PTLDs.
- Composite B-cell and T-cell lineage PTLD: harboring both B- and T-cell clones either concurrently or successively in the same patient is extremely rare and only a few cases have been reported in the literature, exclusively after SOT, with poor outcome.

45.3 Pathogenesis

The pathogenesis of PTLDs is a result of EBV-induced transformation of B-cells in the setting of impaired anti-EBV cellular immunity due to iatrogenic immunosuppression and resulting in an outgrowth of EBV-infected B-cells. GVHD prevention strategies that indiscriminately remove T-cells from the graft increase the risk of PTLD.

Recurrent EBV infection preceding clinically overt PTLD is the consequence of viral latency, which is the stage in the viral life cycle in which no virions are produced, as opposed to the lytic stage. During viral latency of EBV, three associated patterns of viral protein expression, so-called latency programs, may be expressed. During infection of the B-cell, these latency programs guide the B-cell through the germinal center reaction pushing it towards the resting memory cell stage. Different latency proteins are implicated in EBV-driven lymphomagenesis demonstrated by the expression of a particular latency program in different lymphoma subtypes.

B-cells in PTLD express a number of latency proteins which are highly immunogenic and are vigorously targeted by T lymphocytes in immunocompetent hosts. This viral gene program in EBV-PTLD is called type III (infected cells express EBNA1, 2, 3A, 3B, and 3C, EBNA-LP, LMP 1 and 2, EBER 1 and 2, and the microRNAs [miRNA] miR-BHRF1 and miR-BART3) of latency and is different and less immunogenic than in other EBV-related diseases with type I (EBNA1, EBV-encoded small RNA [EBER], and BamHI fragment A rightward transcripts [BART] transcripts) or II (latent membrane proteins, LMP1/2EBNA2, EBNA3, and EBNA-leader protein LP) of latency.

EBV plays also an important role in the pathogenesis and epidemiology of acute and chronic GVHD. As B-cells also play a role in the pathophysiology of chronic GVHD, and B-cells are stimulated into activity by EBV infection and B-cell recovery occurs usually after day +100, the impact of EBV-infected B-cells is stronger for the development of chronic and, to a lesser extent, acute GVHD.

45.4 Clinical Manifestations

Lymphadenopathy and fever are the most common symptoms of EBV-PTLD. Rare EBV-associated PTLD manifestations, also referred to as EBV end-organ disease, include encephalitis/myelitis, pneumonitis, hepatitis, and hemophagocytic lymphohistiocytosis (Table 45.1).

Table 45.1 Clinical manifestations of PTLD

Time to PTLD	• Median time of PTLD development: 2–4 months • 6% PTLD cases are diagnosed within the first month • 90% diagnosed within the first 6 months after HCT • Rarely: >5 years post-HCT (more likely representing lymphoma)
Incidence of EBV-DNA-emia[a]	• Median: 29.4% (range: 0.1–63%) • Highest in MUD/MMUD-HCT, haplo-HCT without PTCy • Lowest in haplo-HCT with PTCy (post-HCT CY)
Incidence of PTLD[a]	• All allo-HCT: 3.22% • MFD-HCT: 1.16% • MMFD-HCT: 2.86% • MUD-HCT[b]: 3.97% • MMUD-HCT[b]: 11.24% • CBT: 4.06% • Auto-HCT without TCD: casuistic
Target organs	• Frequently: lymph nodes • Rarely: CNS, GI tract, lungs, liver

[a] Recent data (during the COVID-19 pandemic) suggest lower incidence of EBV-DNA-emia and EBV-PTLD
[b] Level of donor match determined locally as 8/8 or 10/10

Table 45.2 Diagnostic methods for PTLD

Noninvasive diagnostic methods
• Quantitative determination of EBV-DNA-emia • Imaging: CT or PET-CT (for avid structures: localized in lymph nodes, spleen, liver, GI tract, skin, lungs, bone, BM) or MRI (in CNS disease; and non-avid histologies)
Invasive diagnostic methods
• Biopsy of the lymph node and/or other suspected sites • Endoscopy: when with GI symptoms • Histological examination: (a) Detection of viral antigens or in situ hybridization for EBER (EBV-encoded-RNA) transcripts (b) Immunohistochemistry (c) Flow cytometry for B-cell, T-cell, and plasma cell lineage-specific antigens

45.5 Diagnosis

The diagnosis of EBV-PTLD must be based on symptoms and/or signs consistent with PTLD together with detection of EBV by an appropriate method applied to a specimen from the involved tissue. Definitive diagnosis of EBV-PTLD requires noninvasive and invasive techniques (biopsy and histological examination) (Table 45.2).

PET imaging: By definition, PTLD is a neoplastic lymphoproliferation. Malignant lymphomas have the ability to metabolize 18F-fluorodeoxyglucose (FDG), which is used by PET imaging. In most cases, PTLD has FDG-avid histology; thus, FDG-PET is an important diagnostic tool for this disease (Dierickx et al. 2013; Dierickx et al. 2018).

Monitoring EBV-DNA-emia: Serial quantitative measurement of EBV viral load posttransplant is currently the method of choice for early detection and monitoring progression and response to treatment of EBV-PTLD. Although the value of the viral load in PTLD risk assessment is uncertain, it is recommended to begin the screening in patients with risk factors after hematological recovery and no later than 4 weeks after the day of HCT. In EBV-DNA-negative patients, frequency of screening should be once a week, while in patients with rising EBV-DNA-emia, more frequent sampling might be considered, as the calculated doubling time for EBV might be as short as 56 h. The screening should be continued at least 4 months in high-risk patients. Longer monitoring is recommended in patients considered to have poor T-cell reconstitution, with severe GVHD, after haplo-HCT, with the use of TCD, after conditioning with ATG/alemtuzumab or in those having experienced an early EBV reactivation.

45.6 Risk Factors

Risk of development of PTLD is essentially proportional to the degree of T-cell depletion/impairment and this should be regarded as the principal risk factor. Thus, the type of donor and type of conditioning have secondary value as risk factors. Since in the HCT setting PTLD usually originates from the donor, and the risk of PTLD is obviously higher when the donor is seropositive. Risk factors for PTLD in match family donor (MFD) transplants include TCD ex vivo or in vivo, EBV serology mismatch between donor and recipient, and splenectomy (Uhlin et al. 2014). ECIL-6 classified HCT patients into three groups of the risk for EBV-PTLD: low, standard, and high risk (Table 45.3) (Styczynski et al. 2016b).

Table 45.3 Risk factors for PTLD

Risk group	Patients
High	• MUD/MMUD • Alternative donors including CBT • MFD-HCT with at least one risk factor
Standard	• MFD-HCT without risk factors • Haplo-PTCy-HCT
Low	• Auto-HCT

45.7 Grading

No grading system currently exists for PTLD. It seems that apart from the findings from biopsy material with a histological examination, the diagnostic criteria of tissue involvement in PTLD should be consistent with those for lymphoma (the Ann Arbor and the Lugano classifications). Nowadays, the use of FDG-PET-CT has emerged as an important imaging tool for PTLD diagnosis and staging.

Possible staging of PTLD includes the following:

- Clinical end-organ staging: nodal vs extranodal disease.
- Clinical severity staging: limited (unifocal) vs advanced (multifocal) disease.
- ECIL-6 staging (based on: the Lugano lymphoma classification by PET-CT imaging): limited (stages I-II); advanced forms (stages III–IV).

45.8 Treatment

45.8.1 Prevention: Donor and Recipient Issues

As EBV might be transmitted with the graft, selection of EBV-seronegative donor might be beneficial for EBV-seronegative recipient, if possible. For EBV-seropositive patients, selection of an EBV-seropositive donor might be justified, as transmission of EBV-specific CTLs outweighs the risk of transmission of EBV-positive B-cells from the donor. However, recent data suggest lack of transfer of EBV-CTL from EBV-seropositive donor to recipient.

45.8.2 Treatment Strategies

PTLD has to be regarded as disseminated disease at diagnosis. This is because of the involvement of lymphoid tissue, which localized throughout the whole body. Therapeutic approaches applied in the prevention and treatment of EBV-PTLD include administration of rituximab, reduction of immunosuppression (RIS), use of EBV-CTL, DLI, and chemotherapy, while other methods have only historical value. No antiviral drug is currently effective against EBV. There are three major approaches to EBV infection after HCT: prophylaxis, preemptive therapy (also known as preemptive prophylaxis), and treatment of established EBV-PTLD.

45.8.2.1 Prophylaxis

- Prophylaxis of EBV disease is defined as drug or cellular therapy given to an asymptomatic EBV-seropositive patient to prevent EBV-DNA-emia. This strategy is administered rarely, with the use of rituximab or EBV-CTL. Antiviral drugs are not justified as anti-EBV prophylaxis.
- The rationale for prophylactic use of rituximab before or early after allo-HCT is B-cell depletion. Prophylactic use of posttransplant rituximab reduced the risk of EBV-DNA-emia, with no impact on PTLD incidence, TRM, or OS in comparison to preemptive therapy.
- The prophylactic use of EBV-CTLs resulted in excellent efficacy in patients at a high-risk group for EBV-PTLD. The obstacle for the use of this approach is the limited availability of CTLs in most transplant centers.
- Low risk of EBV-DNA-emia and EBV-PTLD was observed after the use of PT-Cy and sirolimus for GVHD prophylaxis.

45.8.2.2 Preemptive Therapy

- Preemptive therapy denotes drug or cellular therapy given to a patient with EBV-DNA-emia in order to prevent EBV disease.
- Monitoring for EBV-DNA-emia is essential in all patients with risk factors for EBV-PTLD. Significant EBV-DNA-emia without clinical symptoms of disease in high-risk

patients for EBV-PTLD is usually an indication for preemptive therapy.
- The goal of preemptive therapy is to obtain a negative EBV PCR or EBV-DNA-emia below the initial threshold without relapse.
- Usually, EBV-DNA-emia occurs prior to the onset of clinical symptoms. There is a correlation between rising or high EBV-DNA-emia in PB and the development of EBV-PTLD but this not the rule.
- Currently available data does not allow for the determination of an unambiguous EBV-DNA threshold value for the diagnosis of EBV-PTLD or other end-organ EBV disease in HCT patients.
- Apart from the EBV-DNA value, also the kinetics of a rising EBV-DNA-emia, together with an assessment of an individual patient's immune function, are very important when appraising the need for preemptive therapy. Local experience based on the correlation of clinical and laboratory data might be a rationale for center-specific cutoff value.
- The primary method for preemptive therapy includes rituximab, once weekly until EBV-DNA-emia negativity. Usually, 1–2 doses of rituximab are sufficient. Rituximab should be combined with reduction of immunosuppression (RIS), if possible. A contraindication for RIS is severe, uncontrolled acute or chronic GVHD. This approach might have additional benefit of rituximab administration, as rituximab possibly reduces the risk of acute/chronic GVHD. Donor or third-party EBV-CTL is another option, although it is not widely available.

45.8.2.3 Treatment of Established EBV-PTLD

- Treatment of established EBV-PTLD means therapeutic interventions for patients with probable or proven EBV disease. Due to the consequential risk of a rapidly growing high-grade lymphoid tumor, together with the potential for EBV to cause rapid MOF, therapy should be implemented as soon as possible (Styczynski et al. 2013).
- For first-line therapy three options are recommended: (a) rituximab, 375 mg/m^2, once weekly; (b) RIS, if possible, usually together with administration of rituximab; (c) adoptive immunotherapy with cellular therapy with in vitro-generated donor or third-party EBV-CTL, if available.
- For the second-line therapy, in case of rituximab failure the following are recommended: (a) cellular therapy (nonspecific DLI or specific EBV-CTLs, if available) and (b) chemotherapy ± rituximab. Unselected DLI from EBV-positive donor is used in order to restore broad T-cell reactivity, including EBV-specific responses. In 2022, EMA approved tabelecleucel ("off-the-shelf" third-party donor EBV-specific VST, viral-specific T-lymphocytes).
- Not recommended: IVIG, interferon, and antiviral agents should not be used for any anti-EBV therapy of PTLD (Table 45.4).

45.8.2.4 Treatment in CNS Disease

- CNS localization of PTLD is a special form of the disease, due to the risk of neurological consequences even in case of successful eradication of EBV from CNS. No standard therapy has been accepted up to date.
- Possible therapeutic options include the following: (a) rituximab, either systemic or intrathecal (in the latter case, dose of rituximab was 10–30 mg in 3–10 mL saline administered weekly), (b) T-cell therapy with EBV-CTLs, (c) radiotherapy, and (d) chemotherapy ± rituximab according to primary CNS lymphoma protocols based on a high dose of methotrexate ± cytarabine (Czyzewski et al. 2013).

Table 45.4 Response rates to anti-EBV-PTLD therapy

Treatment strategy	Preemptive therapy	Therapy of PTLD
Rituximab	90%	65%
Rituximab + RIS		78%
EBV-CTL	94–100%	71–75%
RIS	68%	61%
DLI		58%[a]
Chemotherapy		26%[a]
Antivirals (cidofovir)		34%[a]

RIS reduction of immunosuppression
[a] With other therapies

45.8.2.5 Criteria of Response to Therapy in EBV-PTLD

- The treatment goal in the EBV-PTLD setting is resolution of all signs and symptoms of PTLD together with negative EBV-DNA-emia.
- The response to rituximab therapy can be identified by a decrease in EBV-DNA-emia of at least 1 log of magnitude in the first week of treatment.
- Positive prognostic factors for outcome to rituximab therapy include age below 30 years, underlying nonmalignant disease, no acute GVHD ≥II, RIS at time of PTLD diagnosis, and decrease of viral load after 1 or 2 weeks of therapy.
- Complete remission of PTLD can be defined as resolution of all symptoms of PTLD, including clearance of EBV-DNA-emia. Partial response of PTLD can be stated with the decrease of at least 50% of initial changes, including decrease of EBV-DNA-emia.
- The response to therapy can be confirmed by achievement of a PET-negative complete remission for avid lymphomas and CT/MRI for non-avid histologies or CNS localization.

Key Points

- Definition: PTLD results from an uncontrolled neoplastic proliferation of lymphoid or plasmacytic cells in the context of extrinsic immunosuppression after transplantation. PTLD in the HCT setting are largely caused by latent EBV. Risk factors for EBV-PTLD are proportional to the degree of T-cell impairment.
- Diagnosis: should be based on invasive techniques including biopsy of the lymph node and/or other sites suspected for EBV disease. Noninvasive diagnostic methods have an accessory value and include the quantitative determination of EBV-DNA-emia in blood, plasma, or serum and PET-CT/CT/MRI.
- Management strategies: prophylaxis, preemptive treatment, and therapy of established EBV-PTLD. Therapeutic approaches include administration of rituximab, reduction of immunosuppression (RIS), use of EBV-CTL (including third-party donor), or DLI and chemotherapy, while other methods have only a historical value.
- EBV-DNA-emia threshold value: No value determines diagnosis of EBV-PTLD or other end-organ EBV disease in HCT patients. In order to initiate preemptive therapy, transplant centers should use own threshold values of EBV-DNA-emia.
- Outcome: Administration of rituximab results in a positive outcome for over 90% of patients treated preemptively and over 65% when it is used as targeted therapy for EBV-PTLD. RIS when applied in combination with rituximab: over 80%. The use of EBV-CTLs: >90% of patients treated preemptively and approximately 75% in therapy of EBV-PTLD.

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